HONOUR-JESUS BEZALEEL

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SUMMARY

Statistics graduate and machine learning researcher specialising in natural language processing for low-resource languages, with experience in speech technology, ASR and TTS evaluation, and multilingual language model assessment. My research interests include robust evaluation metrics for low-resource speech systems, linguistically informed evaluation, and AI safety, particularly cultural alignment in multilingual settings. I aim to advance computationally efficient, context-aware evaluation methods and human-centred multimodal AI systems that improve the reliability and inclusivity of language technologies for under-represented languages and communities, contributing both theoretically and practically to the field.

EDUCATION

B.Sc. Statistics | Ekiti State University, Nigeria *Dec 2019 – Sept 2024*

Relevant Coursework: Probability Theory, Statistical Inference, Linear Algebra, Numerical Methods, Stochastic Processes, and Optimisation.

RESEARCH PUBLICATIONS

- Bezaleel, H.-J. *HausaPQ-Speech: A Validated Multimodal Dataset for Hausa Educational Question Answering.* Manuscript under review.
- Bezaleel,H.-J. Tone on a Budget: A Reference-Free Metric for Lexical Tone in Massively-Multilingual TTS. **Poster, Deep Learning Indaba 2026.**
- Bezaleel, H.-J. (2026). *Why Large Language Models Fail For Hausa Educational Content: Cascading Errors From Translation To Speech To Comprehension.* **WiML@ICML '26**
- Bezaleel, H.-J. (2026). *Why Large Language Models Fail for Hausa Educational Content: Cascading Errors From Translation to Speech to Comprehension.* **ICBINB@ICLR '26.**
- Bezaleel, H.-J. (2025). *Bridging Language Barriers in Education Using TTS–ASR Approaches for Low-Resource Languages in Nigeria.* **Poster, Deep Learning Indaba 2025.**
- Bezaleel, H.-J. (2024). *Statistical analysis of unemployment and job placement in Nigeria.* ResearchGate.
- Bezaleel, H.-J. (2023). Impact of generative AI on beginner-level tech enthusiasts. In Generative AI Possibilities: A White Paper on Use Cases (pp. 12–18).

RESEARCH PROJECTS

Bilingual Hausa–English ASR–TTS System | Low-Resource Speech Research *Aug 2025 – Present*

Manuscript in preparation for submission to EMNLP 2026.

- Designed and implemented a four-stage cascaded NLP pipeline: fine-tuned NLLB-200-distilled-600M for English→Hausa machine translation (trained on a single Tesla GPU under real compute constraints), a fine-tuned VITS model for Hausa TTS synthesis, Whisper-large-hausa for ASR, and Llama-3.3-70B for downstream transcript error correction); built automated data pipelines for dataset creation, preprocessing, and bilingual text–audio alignment.
- Developed evaluation and benchmarking workflows using WER, CER, and MOS, validating performance on the HausaPQ-Speech corpus (1,000 human-validated samples); identified critical performance gaps, including WER variance (68–92%) and 100% tonal error rate in baseline ASR models (Whisper, Wav2Vec2).
- Conducted systematic failure-mode analysis across the pipeline, uncovering key issues such as semantic omission in MT, lexical diversity collapse (TTR: 17.3% → 13.9%), LLM hallucination, and reasoning degradation (82% → <30% accuracy) in downstream tasks.
- Optimised pipeline performance for real-world multilingual deployment, replacing closed-source MT with fine-tuned NLLB-200 for reproducibility and implementing structural normalisation and inference stabilisation techniques in compute-constrained environments.

Self-Training with Confidence-Based Filtering for African ASR at Scale | *Jan 2026 – Present*

Submitted / under review.

- Designed a multi-signal pseudo-label filtering framework combining decoder log-probability, CTC-decoder agreement (auxiliary CTC head on Whisper encoder), length-ratio constraints, and language ID confidence, retaining only samples passing all four gates.
- Engineered the CTC agreement component: attached a CTC head to the Whisper encoder and computed token-level agreement between CTC and autoregressive decoder outputs to flag incorrect transcriptions.

- Designed the full ablation study across seven filter combinations and four unlabelled data scales (25h–200h per language) to isolate per-signal contribution and identify filtering saturation points.

Geo-NLP Solutions for Food Insecurity Analysis | *Apr 2024 – Aug 2024*

- Built a decision-support system combining NLP and geospatial analytics to analyse 1,000+ reports, reducing manual analysis time from weeks to hours.
- Applied clustering techniques (HDBSCAN, K-Means) to identify high-risk regions and hidden patterns in food insecurity data.
- Developed an interactive geospatial dashboard (GeoPandas, Folium) used by stakeholders to guide resource-allocation decisions.

EXPERIENCE

AI/Machine Learning Engineer (Freelance & Researcher) | Remote *Oct 2023 – Present*

- Built an end-to-end API that converts research papers into structured audiobooks: integrated PDF parsing, local LLM inference, and TTS to handle equations, tables, and citations; containerised with Docker and deployed on AWS/GCP; evaluated with ROUGE, BLEU, and WER. [GitHub link]
- Designed and deployed an automated model benchmarking and selection platform for Hugging Face models using FastAPI and Gradio, enabling multi-criteria evaluation across latency, memory usage, and task-specific performance on AWS EC2 infrastructure. [GitHub]
- Engineered preprocessing and alignment infrastructure for low-resource neural machine translation systems supporting English → Kalenjin and Kinyarwanda transformer-based pipelines, seq2seq models; engineered preprocessing workflows covering tokenisation, alignment, and normalisation. [GitHub]

Machine Learning Researcher | Data Science Nigeria *Aug 2023 – Oct 2023*

- Co-led development of a speech-to-text financial advisory chatbot using NLP methods, deployed to 100,000+ users across Nigeria and improving access to financial knowledge in underserved communities.
- Engineered Geo-NLP pipelines to extract contextual insights from heterogeneous geospatial and textual data, collaborating with a cross-disciplinary team of geospatial analysts and ML researchers.
- Designed a GIS-based visualisation framework that mapped customer distributions and surfaced patterns in financial access, directly informing evidence-based financial inclusion strategy.
- Contributed to family planning dataset curation and annotation, ensuring data quality and consistency for downstream model training.

Data Scientist Intern | OneCampus Academy *Mar 2022 – Jun 2023*

- Built and deployed a credit risk classification model in Python (Scikit-learn) that improved loan servicing efficiency by 20%, supporting analysts in real-time approval decisions.
- Engineered a SQL pipeline joining 5+ tables across 200,000+ customer records, uncovering repayment behaviour patterns that lifted model precision by 12%.
- Developed and benchmarked a fraud detection system using PyTorch RNNs, achieving a 23% accuracy gain over baseline classifiers and measurably reducing financial exposure.
- Delivered a customer segmentation tool (Python, Streamlit) enabling targeted product recommendations and reducing churn risk.

Data Science Intern | Hamoye. com *Jan 2022 – Apr 2022*

- Designed and deployed a time-series forecasting model that improved planning accuracy by 15%, enabling more reliable operational decision-making.
- Implemented reproducible data pipelines for preprocessing and model training across structured and unstructured datasets.
- Collaborated with analysts to translate data findings into actionable business insights.

TECHNICAL SKILLS

Languages	Python (Pandas, NumPy, Scikit-learn, Matplotlib), SQL (PostgreSQL, MySQL), R
ML / DL	PyTorch, TensorFlow, Keras, Scikit-learn, supervised/unsupervised learning, optimization, time-series forecasting
NLP & Speech	HuggingFace Transformers, spaCy, LangChain, LLM fine-tuning, ASR/TTS, seq2seq
MLOps / Infra	Docker, FastAPI, Flask, AWS (EC2, S3), GCP, Azure, Git, inference optimisation
Evaluation	WER, BLEU, ROUGE, CER, MOS, benchmark design, and model failure analysis
Visualisation	Power BI, Plotly, Streamlit, GeoPandas, Folium